

Practical 7

Task 1

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
#include <stdio.h>
#include <stdlib.h>
/* Global variable */
int global_var = 10;
/* Static global variable */
static int static_global = 20;
/* Function - Code/Text segment */
void display()
{
    printf("This is a function\n");
}
int main()
{
    /* Stack variable */
    int stack_var = 30;
    /* Static local variable */
    static int static_local = 40;
    /* Heap variable */
    int *heap_var = (int *)malloc(sizeof(int));
    *heap_var = 50;
    printf("----- MEMORY ADDRESSES ----- \n");
    /* Code segment */
    printf("Code address          : %p\n", (void *)display);
    /* Global segment */
    printf("Global variable       : %p\n", (void *)&global_var);
    /* Static global */
    printf("Static global          : %p\n", (void *)&static_global);
    /* Static local */
    printf("Static local          : %p\n", (void *)&static_local);
}
```

OUTPUT:

```
root@Tejasri:~# nano Memory.c
root@Tejasri:~# gcc Memory.c -o Memory
root@Tejasri:~# ./Memory
----- MEMORY ADDRESSES -----
Code address          : 0x60421a4881c9
Global variable       : 0x60421a48b010
Static global         : 0x60421a48b014
Static local          : 0x60421a48b018
Heap variable         : 0x6042593a02a0
Stack variable        : 0x7ffec2eb98dc
root@Tejasri:~#
```

Task 2

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
#include <stdio.h>
#include <unistd.h>
int main()
{
    FILE *file;
    char filename[50];
    char line[256];
    /* Get process ID */
    pid_t pid = getpid();
    printf("Process ID: %d\n", pid);
    /* Create /proc/PID/maps path */
    sprintf(filename, "/proc/%d/maps", pid);
    /* Open memory map */
    file = fopen(filename, "r");
    if (file == NULL)
    {
        perror("Unable to open /proc/PID/maps");
        return 1;
    }
    printf("\n----- PROCESS MEMORY MAP ----- \n\n");
    /* Read and display each line */
    while (fgets(line, sizeof(line), file))
    {
        printf("%s", line);
    }
    fclose(file);
    return 0;
}
```

OUTPUT;

```
root@Tejasri:~# nano proc.c
root@Tejasri:~# gcc proc.c -o proc
root@Tejasri:~# ./proc
Process ID: 593

----- PROCESS MEMORY MAP -----

613c6dcd3000-613c6dcd4000 r--p 00000000 08:30 36948 /root/proc
613c6dcd4000-613c6dcd5000 r-xp 00001000 08:30 36948 /root/proc
613c6dcd5000-613c6dcd6000 r--p 00002000 08:30 36948 /root/proc
613c6dcd6000-613c6dcd7000 r--p 00002000 08:30 36948 /root/proc
613c6dcd7000-613c6dcd8000 rw-p 00003000 08:30 36948 /root/proc
613ca2b54000-613ca2b75000 rw-p 00000000 00:00 0 [heap]
70a81ce00000-70a81ce28000 r--p 00000000 08:30 18399 /usr/lib/x86_64-linux-gnu/libc.so.6
70a81ce28000-70a81cfb0000 r-xp 00028000 08:30 18399 /usr/lib/x86_64-linux-gnu/libc.so.6
70a81cfb0000-70a81cff0000 r--p 001b0000 08:30 18399 /usr/lib/x86_64-linux-gnu/libc.so.6
70a81cff0000-70a81d003000 r--p 001fe000 08:30 18399 /usr/lib/x86_64-linux-gnu/libc.so.6
70a81d003000-70a81d005000 rw-p 00202000 08:30 18399 /usr/lib/x86_64-linux-gnu/libc.so.6
70a81d005000-70a81d012000 rw-p 00000000 00:00 0
70a81d163000-70a81d166000 rw-p 00000000 00:00 0
70a81d16b000-70a81d16d000 rw-p 00000000 00:00 0
70a81d16d000-70a81d171000 r--p 00000000 00:00 0 [vvar]
70a81d171000-70a81d173000 r--p 00000000 00:00 0 [vvar_vclock]
70a81d173000-70a81d175000 r-xp 00000000 00:00 0 [vdso]
70a81d175000-70a81d176000 r--p 00000000 08:30 18396 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70a81d176000-70a81d1a1000 r-xp 00001000 08:30 18396 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70a81d1a1000-70a81d1ab000 r--p 0002c000 08:30 18396 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70a81d1ab000-70a81d1ad000 r--p 00036000 08:30 18396 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70a81d1ad000-70a81d1af000 rw-p 00038000 08:30 18396 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7fff9e636000-7fff9e657000 rw-p 00000000 00:00 0 [stack]
root@Tejasri:~#
```